Atmo-Ensemble
Ensemble ML Research Conclusion for Air Quality Multi-Classification

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Abstract

1 Baselines and Exploratory Data Analysis

1.1 Dataset Import and Overview

The analysis begins with the extraction and importation of the Air Quality and Pollution Assessment dataset. Initial data exploration confirms that the dataset contains 5,000 samples and 10 features, with a perfectly clean state: there are no missing values (0 missing cells) and no duplicated rows.

The variables measure different meteorological metrics such as, pollution concentrations, and geographic factors: Temperature, Humidity, $PM_{2,5}$, PM_{10}, NO_2, SO_2, CO, Proximity to Industrial Areas, and Population Density.

1.2 Target Column Distribution and Encoding

The target variable, Air Quality, is imbalanced but follows a clear structured distribution:

- **Good**: 2,000 samples (40.0%)
- **Moderate**: 1,500 samples (30.0%)
- **Poor**: 1,000 samples (20.0%)
- **Hazardous**: 500 samples (10.0%)

In order to prepare the data to work only with numerical algorithm, the target column was encoded into numerical values between 0 and 3 (Hazardous = 0, Poor = 1, Moderate = 2, Good = 3).

1.3 Feature Distributions by Class

The average feature profile of each class was evaluated using standardised values, visualised through a heatmap and a parallel coordinates plot.

A clear monotonic gradient is visible across all pollution features:

- `Good` conditions sit clearly above the dataset mean.
- `Hazardous` conditions sit firmly at the bottom of the plot.
- `Poor` and `Moderate` follow the exact same pattern, being in between those extreme values.

The only exception to this trend is Proximity_to_Industrial_Areas, which completely changes the pattern. Since a greater distance from industry usually belongs to cleaner air, this variable shows an inverse relationship compared to the other more direct pollution measurements.

On top of that, as we can see in the Feature Importance cell, CO2 and Proximity to industrial Areas have respectively 0.35 and 0.29 percent of weight in the definition of the target column. Therefore, in order to test sensor failure, we have randomly dropped CO2 metrics.

Additionally, The heatmap and parallel coordinates plot show the average feature profile of each class using standardised values. A clear monotonic gradient is visible across all pollution features, `Hazardous` sits clearly above the dataset mean while `Good` sits in the bottom of the plot. With `Poor` and `Moderate` following the same pattern in between those extreme values. The only exception is `Proximity_to_Industrial_Areas`, which completely changes this pattern, as more distance from industry usually corresponds to cleaner air. The parallel coordinates plot reflects this clearly, showing four well-separated lines for every feature, with the exception of the `Proximity_to_Industrial_Areas` marking the highest confusion risk zone. Additionally, 'Moderate' is almost flat near zero across every feature, meaning its centroid sits close to the dataset average with little changes, meaning that it is the hardest class to separate from its neighbours.

1.4 PCA and Dataset Feature Extraction

Does studying correlated environmental features through principal component analysis (PCA) and other feature extraction techniques improve the robustness and efficiency of ensemble classifiers, or are non-linear feature interactions found by the models alone?

To understand if dimensionality reduction benefits our predictive modelling, we have evaluated a pipeline combining Principal Component Analysis (PCA) with a Standard Scaler and a Random Forest Classifier. We have tested the model's performance, we have used Macro F1 and 5-fold cross validation to measure the results, across a varying number of PCA components, from 2 to 9.

The results have shown that the maximum performance with an F1-score of 0.9050 was achieved when we kept 9 components, which means that is better retaining the full dimensionality of the dataset we have. While a local performance maximum was observed at 5 components, the general trend indicated that PCA compression does not significantly improve the model's predictive ability.

To summarize, for ensemble classifiers like the one we used, Random Forests, PCA compression does not offer any clear benefits in this context. The model performs better when it uses all the original components, meaning that tree-based ensembles are capable of discovering complex interactions directly from the raw data. As a result, we have maintained the original feature space, but it also implies that the model's overall metrics could suffer severe changes when simulating real-world sensor errors or noise on those raw features.

2 Research Conclusions

2.1 Ensemble Learning

Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?

2.1.1 Overview of Evaluated Models

To address this task, we built different ensemble learning based models and evaluated them to classify air quality into four categories: Good, Moderate, Poor, and Hazardous. The main focus was getting a final model able to correctly predict pollution. The evaluated models include:

- **Voting Ensembles**: Hard voting (majority) and Soft voting (average probability) combining Logistic Regression, Random Forest, and Support Vector Classifier.
- **Parallel Ensembles (Variance-Reduction)**: Bagging (with replacement) and Pasting (without replacement) utilizing parallel training on data subsets.
- **Sequential Ensembles (Bias-Reduction)**: Boosting methods like AdaBoost and Gradient Boosting, which use sequential training to correct prior errors.
- **XGBoost**: A high-performance gradient boosting implementation featuring system optimization and regularization.
- **Stacking Ensemble**: A multi-layer ensemble using a final meta-learner to blend predictions.

Because of the class imbalances in the dataset, special importance was given to the Macro F1-score, but other metrics where used to understand the situation: Accuracy, Weighted F1-Score, and the Matthews Correlation Coefficient (MCC).

2.1.2 Parallel vs. Sequential Ensembles Performance

The parallel tree ensembles, Bagging and Pasting, gave us strong variance reduction, achieving Macro F1-scores of 0.9240 and 0.9204, respectively. Their out-of-bag (OOB) score of 0.9504 provided a trustable estimation of models performance. However, sequential ensembles, which focus on correcting the errors of previous models, outperformed them.

While the fine-tuned AdaBoost model reached a Macro F1 of 0.9209, Gradient Boosting achieved a higher Macro F1 of 0.9297, surpasing the parallel methods. Finally, XGBoost obtained the highest overall scores with an Accuracy of 0.9560, a Macro F1 of 0.9351, and an MCC of 0.9371.

2.1.3 Key Insights and Conclusion

Based on the obtained results, the bias-reduction approach of sequential ensembles offered better generalization on the sensor data compared to the variance-reduction of parallel ensembles. While bagging offered robustness, the boosting methods, particularly XGBoost, brought the best predictive performance.

Despite the strong overall performance, a class imbalanced represented a key challenge for the task, specifically for the detection of the minority class. All models resulted on lower recall values for this minority class, though XGBoost detected the highest proportion of hazardous cases with a recall of 0.8712.

This difficulty in detecting severe pollution cases highlights the need for other machine learning techniques, such as cost-sensitive learning, previous to a real-world deployment ready model. Lastly, XGBoost was selected as the best model and saved as the final model for this task.

2.2 Handling Ordinality

Does encoding ordinality via Regressor-based Ensembles outperform standard Multi-class Classifiers in Air Quality Assessment?

Our target variable — Good, Moderate, Poor, Hazardous — is not a nominal class. It carries an inherent order:

$$\text{Good} < \text{Moderate} < \text{Poor} < \text{Hazardous}$$

Standard multi-class classification ignores this structure entirely, treating a prediction of Good when the true label is Hazardous as an error equivalent in severity to predicting Moderate. In a real-world air quality context, these are fundamentally different mistakes.

Approach	Description
Standard Multi-class	Treats the 4 classes as independent nominal labels. No ordering assumed.
Frank & Hall	Decomposes the problem into 3 binary classifiers: $P(y > \text{Good})$, $P(y > \text{Moderate})$, $P(y > \text{Poor})$. Outputs are combined to recover the 4-class distribution while preserving ordinal structure.
Regressor Ensemble	Trains an XGBoost Regressor on the numeric labels $\{0, 1, 2, 3\}$, then maps the continuous output back to the nearest class via tuned thresholds.

All three models are fine-tuned with Optuna + MedianPruner. Evaluation uses both standard metrics (Accuracy, F1, MCC) and ordinal-aware metrics that penalise predictions proportionally to how far off they are.

2.2.1 What the metrics tell

The comparison uses both standard metrics like Accuracy, F1, and MCC, and ordinal-aware metrics that measure the magnitude of errors, not just not just if a prediction is predicting the right or wrong value

The key metrics to focus on are:

- **QWK** (Quadratic Weighted Kappa): the primary ordinal metric; higher values mean errors are concentrated near the diagonal.
- **MAE**: lower is better; tells you on average how many steps away predictions land from the true label.
- **Severe Error Rate**: the most safety-critical metric; a `Good` predicted as `Hazardous` is a worse mistake than a `Good` predicted as `Moderate`.
- **Adjacent Accuracy**: an ordinal alternative to accuracy. A prediction that is off by one class still scores.

2.2.2 What the results show

All three models achieve extremely high overall accuracy (between 94% and 95%) with no severe errors on the clean dataset, which reflects a synthetic nature of the data, which tells us that class boundaries are well-separated in the original feature space.

The ordinal methods do not outperform the standard classifier by a large margin on clean data, which is mainly what we expected. Because classes are easily separable, there is little ordinal structure to exploit.

2.2.3 Frank & Hall

The monotonicity violation rate is just 0.10%, which means the binary classifiers are producing consistent probabilities almost every time. The ordinal decomposition assumption holds well on this data, so the enforcement step is rarely needed.

2.2.4 Regressor Ensemble

The regressor takes a different approach, instead of decomposing the problem into binary decisions, it learns a continuous pollution score and uses Optuna-tuned thresholds to map it back to classes. This makes it more flexible and less dependent.

2.2.5 Conclusion

On clean data the three models are essentially tied, but the **Frank & Hall** being a little bit better. The question is what would happen if the data gets harder, if we have more examples or if the values were more overlapped. Would ordinal methods hold up better under class overlap? That is what we have done in the next part of the notebook.

2.2.6 Robustness Analysis

To investigate the effect of class overlap, isotropic Gaussian noise at varying levels $\sigma \in \{0.0, 1.0, 1.5, 3.0\}$ is injected into the feature space, and all three models are re-evaluated at each noise level.

2.2.7 Results across noise levels

As we can see in the results, as noise increases all three models perform worse, dropping from 94-95% accuracy at $\sigma = 0$ down to 83–85% at $\sigma = 3.0$. That part is expected and nearly the same across all the models and different noise levels. The more interesting differences show up in the ordinal metrics.

At $\sigma = 1.0$, all three models still produce zero severe errors, and performance is very close to the noise-less models QWK around 0.95 with Standard Multiclass slightly ahead (0.9552 vs 0.9539 and 0.9534). No method is clearly better at this noise level, as it is the most close one with the original data.

At $\sigma = 1.5$, Frank & Hall starts to be the best at every metric, but not really ahead of the other models. The ordinal models are the only ones with zero severe errors, and F&H leads on every ordinal-based metric. This seems to be the noise level where encoding ordinality starts to affect the results, but without a clear winner.

At $\sigma = 3.0$, as we expected, all the metrics for the 3 models are worse, but still quite good. All three models produce the same severe error rate, so the ordinal advantage disappears under heavy noise. Frank & Hall still leads slightly on QWK and MAE, but the gaps are still small. The Regressor Ensemble is the weakest model of the three at this high noise level.

2.2.8 Conclusions

To summarise, on clean or little noisy data the three models are essentially almost the same, as they produce really high, but still similar results. At moderate noise Frank & Hall is the most conservative, it avoids severe errors longer than the standard classifier and the regressor. At very high noise all methods converge and the ordinal advantage goes away. The Regressor Ensemble, despite being, in theory, the most flexible, does not outperform the other two under noise on this dataset. We also expected the results to be worse as we increased the noise level, but it hasn't been a clear winner between the three models. As a last conclusion, we can say that both three models have learnt the mathematical formula behind the classification of the air quality level, so there is not a clear outcome of which method is a better approach to this problem as well as there isn't a clear finding on if taking into account the ordinality matters.

2.2.9 Future Work

The following directions could deepen the analysis:

- **Using real-world data**: validate on a real air quality dataset where class overlap exists naturally rather than being artificially injected, with real outliers, missing values, and realistic sensor characteristics.
- **Alternative ordinal methods**: explore frameworks with stronger theoretical guarantees such as CORAL or proportional odds models as additional baselines.
- **Cost-sensitive training**: explicitly penalise severe errors during training via a custom loss function, rather than only measuring them after all the process.
- **Threshold refinement**: investigate per-class or per-region threshold tuning for the Regressor Ensemble, since the optimal separation between adjacent classes may

not be uniform across the feature space.

- **Realistic noise**: replace isotropic Gaussian noise with feature-specific perturbations that reflect actual sensor degradation or miscalculation patterns.

Ultimately, explicitly modelling the ordinal nature of air quality labels via the Frank & Hall method yielded the most robust and safety-conscious predictive performance.

2.3 Imbalanced Learning

Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?

The goal of this section is to manage the class imbalance in our air quality dataset, where some categories are less represented.

2.3.1 Managing Class Imbalance

An initial evaluation of the dataset's distribution confirmed a significant class imbalance. The critical hazardous class was underrepresented compared to other classes, making up for only 10% of the samples. In contrast, the majority classes dominated the dataset, with the good category at 40%, moderate at 30%, and poor at 20%. This imbalance gave us the need for special approaches to ensure rare but dangerous air quality samples are not ignored by the models.

2.3.2 Evaluation of Ensemble Methods

Standard ensemble architectures, such as a baseline Random Forest, were compared against specialized methods prepared for imbalanced data, including Balanced Random Forest and RUSBoost. Imbalanced-data approaches showed different results:

while they generally focused more on the minority class, aggressive undersampling in RUSBoost brought a reduction in overall performance.

SMOTE-Boost struggled to maintain high accuracy on the majority classes too.

Lastly, the Balanced Random Forest appeared to be the strongest model in this task for accurately detecting the rare hazardous cases without sacrificing too much performance overall.

2.3.3 Algorithmic Modifications

To further address the imbalance, a parallel approach to data resampling was tested using algorithmic modifications via cost-sensitive learning.

This involved applying severity-based penalties for misclassifications, heavily penalizing the model for missing severe air quality events. The cost-sensitive Random Forest made good use of these penalties, successfully boosting its "Hazardous" detection rates. However, despite these improvements, it still remained behind the Balanced Random Forest in overall effectiveness.

2.3.4 Robustness Testing for Missing Data

To simulate realistic real-world sensor failure conditions, a 30% sensor dropout was introduced to evaluate how well the algorithms could handle missing values (NaNs).

In this scenario, XGBoost showcased a massive advantage due to its capacity to handle missing data. Even with the 30% data dropout, XGBoost maintained a recall rate for the hazardous class of over 90%, outperforming standard Random Forest pipelines that relied on traditional data imputation strategies by far.

2.3.5 Conclusion and key insights

The research concluded that Balanced Random Forest and XGBoost kept a good proposal for real-world air quality monitoring AI solutions. Balanced Random Forest worked best when put against hard class imbalance situations, being able to correctly predict under-represented classes. Meanwhile, XGBoost guaranteed reliable performance, robustness, and generalization when realistic sensor dropouts and missing data happen. Depending on the situation and the use case, these models could provide the necessary safety and resilience required for functional monitoring systems where detecting rare events and handling noisy data are equally relevant.

2.4 Semi-Supervised Learning

Can semi-supervised learning reduce labeling costs while maintaining strong classification performance in air quality prediction?

The primary goal of the experiment was to explore ways to reduce the time and cost associated with manually labeling a complete dataset by evaluating semi-supervised learning techniques. Specifically, the tasks included:

- **Simulating Unlabeled Data:** We partially deleted the target feature (Air Quality) in different proportions to mimic a real-world scenario where only a limited labeling budget is available.
- **Comparing Masking Strategies:** We implemented two different strategies for simulating unlabeled samples: Pasive and Active masking.
- **Passive Semi-Supervision:** Pseudo-randomly masking target samples without taking class distribution into account.
- **Active Semi-Supervision:** Masking samples in a way that leaves the same amount of labeled samples from each class, respecting the original distribution.
- **Evaluating Models:** We ran active and passive semi-supervised learning processes using models like Label Spreading and Self-Training (with a Random Forest base classifier). The unlabeled samples were treated as the test set to check how well the models could self-train and predict the missing labels compared to a supervised baseline.

The experiment gave us several key insights regarding the viability of these semi-supervised methods:

- **Self-training was consistently the strongest approach** comparing it to label spreading across all ratios, specially when labeled data was low (1 to 10%), reaching the best accuracy and macro F1 in almost every train/test distribution.
- **Passive masking usually matched or outperformed active masking** for self training, this tells us that mantaining the original proportions in labeled samples worked well for our problem.
- **Active masking helped label spreading in low labeled sample proportion sets**, but this advantage was not mantained as the labeled sample ammount increased, still underperforming against self training.
- **More labeled data generally improved all methods**, but we need to take into account that those cases leave us with smaller test sets, with less samples to miss, and probably the majority of the set being part of the majority class, but good scores in macro f1 tell us that if a test sample was part of the minority class, it was correclty predicted too, therefore it worked well overall, even if we should take those metrics carefully.

If the labeling budget is limited, using just 30% to 50% labeled data already provides strong model performance. Consequently, full dataset manual labeling can be avoided with only a minimal loss in accuracy and F1 score.

2.5 Efficiency vs. Accuracy in Edge Use Cases

Can lightweight ensembles replace deep stacking for edge-based Air Quality Monitoring?

2.5.1 Trade-offs between Computational Cost and Classification Performance

The primary goal of the experiment was to evaluate the trade-offs between computational cost, model size, and classification performance in the context of "edge machine learning." Because air quality monitoring is typically performed on small, low-power edge devices in the city, deploying a massive ensemble model could be impractical for battery-powered chips.

The tasks included:

- **Comparing Heavy vs. Light Models:** We aimed to find the frontier between model size and performance by comparing a "Heavy Model" (a 3-layer Stacking Classifier with XGBoost, Random Forest, and SVM) against a "Light Model" (a highly pruned Decision Tree).
- **Measuring Operational Metrics:** Rather than relying solely on traditional metrics like Accuracy and F1-Score, we explicitly measured Inference Time (in milliseconds) and Model Size (in Megabytes), as they are values that have real weight when designing an edge AI soution.
- **Addressing Uncertainty (Conformal Prediction):** Since predicting air quality incorrectly (e.g., classifying "Hazardous" air as merely "Moderate") can lead to serious health problems, the notebook evaluated how certain the model was about its predictions. Using MAPIE (Model Agnostic Prediction Interval Estimator), we dealt with critical errors by computing coverage and average prediction set sizes for the different classes.

The experiment provided crucial insights into both the deployment viability and the reliability of the models:

- **Efficiency vs. Accuracy Trade-off:** While the Stacking Classifier (Heavy Model) achieved higher Accuracy (~0.95) and Macro F1-scores (~0.93) than the optimized Decision Tree (~0.91 Accuracy, ~0.87 Macro F1), the Heavy Model was severely penalized in operational cost. It was over 826 times larger in file size (MB) and more than 120 times slower overall, during inference time compared to the lightweight Decision Tree.
- **Dangerous Critical Class Coverage:** During the conformal prediction analysis, the model exhibited less than 0.90 coverage on the critical "Hazardous" class. Specifically, the true label was missing from the prediction set roughly 32% of the time, which is considered a dangerous failure rate for the most health-critical class.
- **Empty Prediction Sets:** The analysis revealed an average set size of less than 1 (or 100%), meaning that some samples were receiving empty prediction sets where no class was included at all, an anomaly that should never occur in a real world scenario.

2.5.2 Conclusions and Key Insights

As we expected, the heavy model took more time to draw a conclusion comparing it to the decision tree, but the target class is not correctly predicted by the smaller model in the specific sample randomly selected to test, even if it only misses the grade of the air toxicity by 1. In a real world scenario, where the minority class is the most dangerous one, missclassifying it, even if only by one space difference, can cause enormous damages, even if the results are impresive given the size difference of 17kB against 13MB. Depending in the hardware, the smaller model could do the job, but if the use case required it, the heavy model would bring more accurate results, given we optimized the hyperparameters to take the minority class more into account (with the Macro F1 Score focused optuna).

In the case of joining classes in the predictions by conformal prediction techniques, the models demonstrated good softmax values in the majority classes, but failed to output reliable values for the minority classes (where the system was previously thought to potentially benefit from joining classes).

Therefore, even if the models had small issues when predicting the minority class, the misclassifications being just 1 class appart from the true value, we believe that these are great results given the weights and the inference time of the models, heavy and small, and depending on the hardware used for deployment, we could select the most suitable one.

3 Ethical and Social Implications

The main purpose of this project is to support safer and more informed air quality monitoring. In that sense, the potential social value is significant: a reliable classifier can help public safety, environmental agents, and citizens overall, supporting faster reactions to dangerous pollution events. However, as this task is directly related to human health, the model must be treated as a decision-support tool rather than a replacement for expert judgement.

3.1 Risk Assessment and Public Health

The most important ethical risk in this problem is a false negative on the `Hazardous` class. Misclassifying dangerous air as `Moderate` or `Good` could turn out to protective actions, expose vulnerable populations to the danger of industrial toxines, and reduce trust in automated systems. For this reason, the project prioritised Macro F1 and other minority-class focused metrics, rather than accuracy alone.

Because of this situation, the analysis is emphasised on severe error rate, ordinal consistency, robustness under noise, class imbalance, and missing sensor readings. In practice, a model used in deployment should be calibrated with caution, especially when used to predict classes near to the hazardous boundary.

3.2 Fairness, Reliability, and Dataset Limitations

Although the dataset is clean and well-structured, it may not fully represent the complexity of real-world environments. Real monitoring systems can be affected by seasonal effects, sensor drift, regional differences, maintenance issues, and unobserved variables such as weather patterns or local emissions spikes. As a result, strong performance on this dataset does not promise equivalent behaviour in operational settings.

Another limitation is that the class distribution is highly imbalanced. Even though the models were evaluated with metrics that take imbalanceness into account, the minority class remains harder to understand and predict. This creates an ethical challenge: a model can appear highly accurate on paper while still failing on the most critical cases. For this reason, any future deployment should include additional checks prior to real integration, such as cost-sensitive training, calibration checks, and continuous monitoring of minority-class recall.

3.3 Final Ethical Reflection

Overall, this work can contribute positively to environmental monitoring and public safety, but only if it is used carefully. The ethical objective is not simply to maximise predictive performance, but to minimise harmful mistakes, especially in the hazardous class.

4 Conclusion Summary

This project explored management of imbalanced datasets on ensemble models for air quality prediction tasks, from multiple machine learning perspectives. Across the different experiments, the dataset proved to be highly structured and relatively easy to segment, which led to strong overall results for most models.

The exploratory analysis showed clear class-dependent patterns we were able to find, with strong monotonic trends across the target categories. PCA did not improve performance, suggesting that tree-based ensembles were already able to exploit the full raw feature space effectively. In the ensemble comparison, sequential methods outperformed parallel ones, and XGBoost achieved the best overall balance of Accuracy, Macro F1, and MCC. This made it the final selected model for the main classification task.

When ordinality was explicitly modelled, the Frank & Hall approach was the most consistent ordinal method, although the advantage over standard multi-class learning was minimal on clean data. Under increasing noise, ordinal modelling showed some benefits at moderate perturbation levels, but the three approaches got to the same level at the end.

For imbalanced learning, Balanced Random Forest and XGBoost stood out as the most useful approaches for ensuring minority-class focused performance. In semi-supervised learning, self-training performed best, showing that a limited annotation budget can still produce strong results. In the efficiency study, the lightweight tree model was far more practical for edge deployment on embedded systems, but the heavier stacking model achieved better predictive quality, more suitable for more powerful devices. Conformal prediction also revealed an important limitation: uncertainty handling for the hazardous class was not yet reliable enough for safety-critical use, therefore, not a suitable technique for our task.

In summary, the project shows that ensemble machine learning techniques can model air quality effectively, but the best approach depends on the goal. If maximum predictive performance is required, XGBoost is the strongest option. If deployment restrictions represent a great limitation, a lightweight model may be preferable, although at some cost in accuracy. Future work should focus on more complete datasets, at least mantaining a minimum ratio of 10% on the minority class, stronger ordinal methods, cost-sensitive learning, and more realistic noise and missing-data scenarios before any practical deployment.